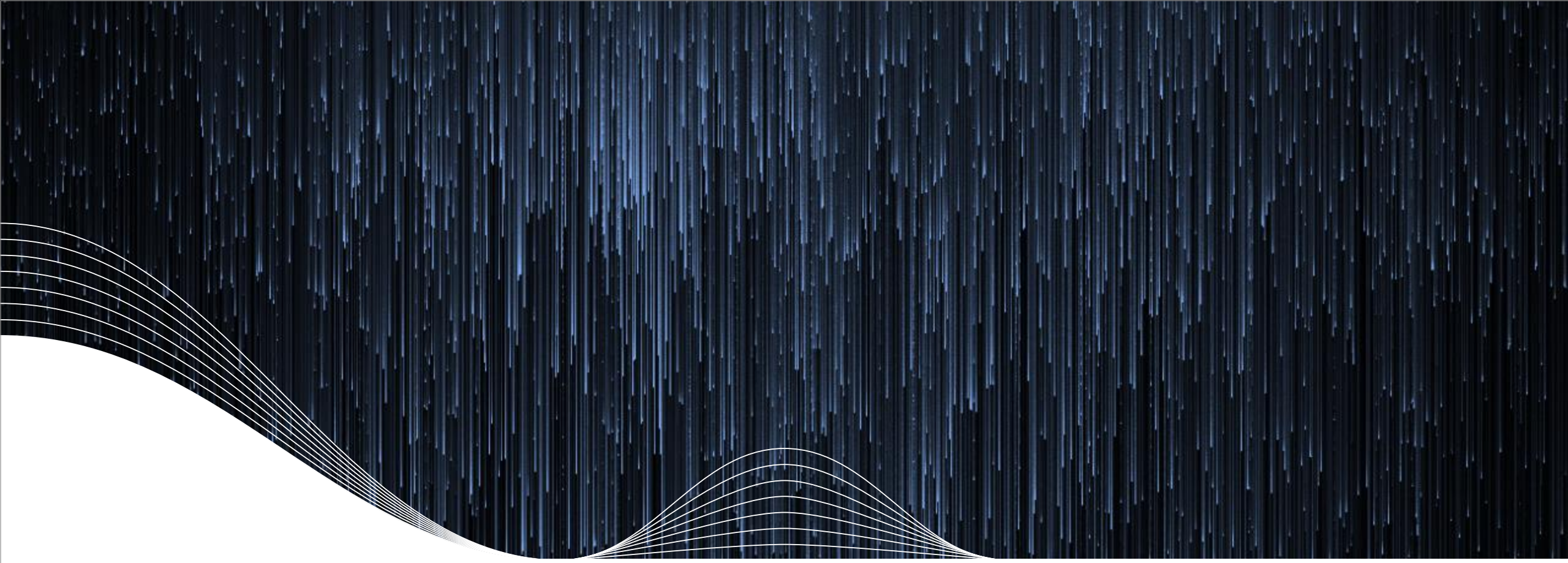


Preparing for a quantum-safe future

Securing the Law Firm

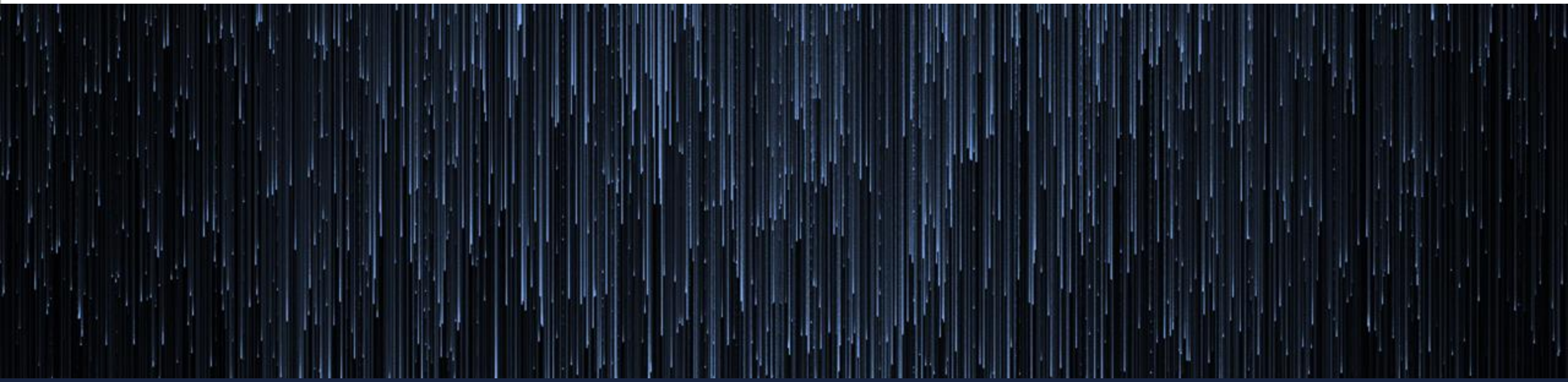
January 2025

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Quantum Leap

Preparing for a quantum-safe future

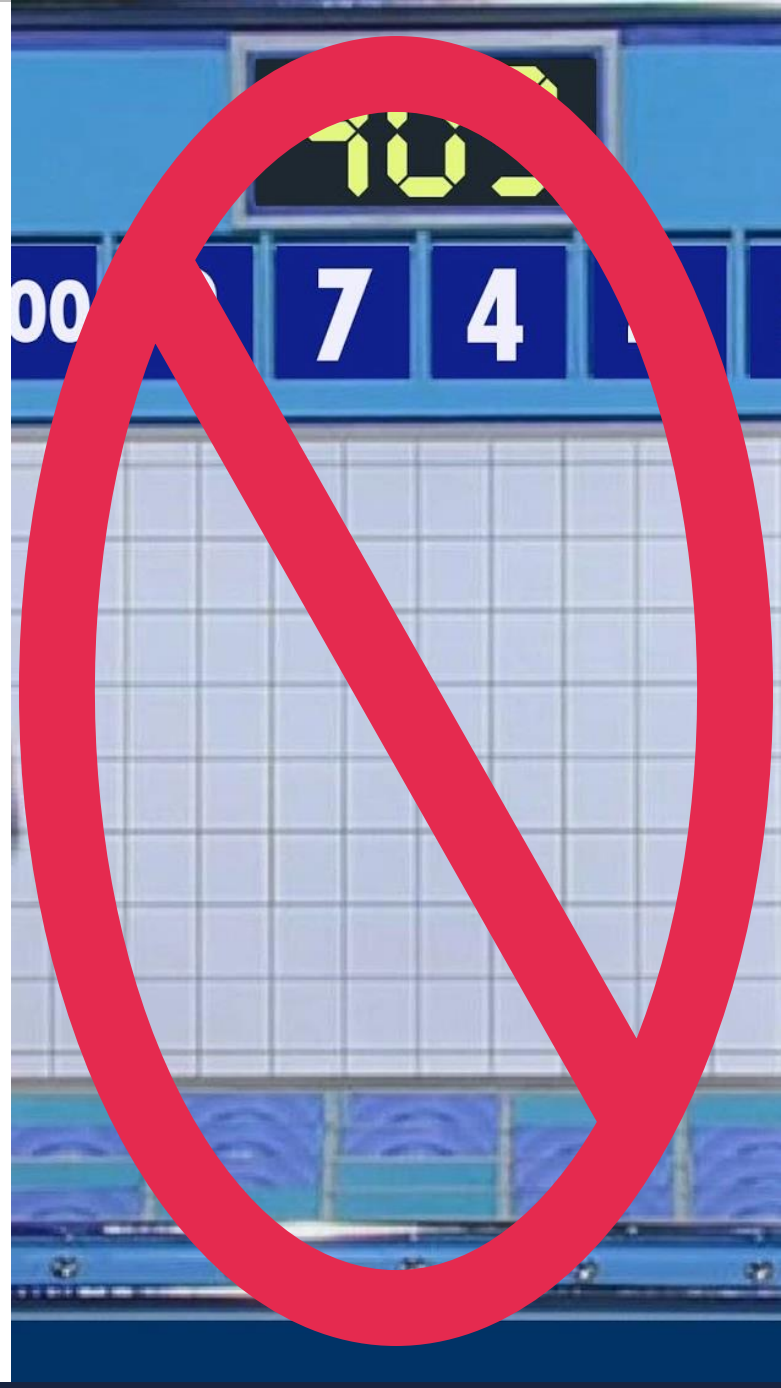
- What is quantum computing and what does it mean for the enterprise?
- What are the risks and how serious is the threat from quantum computing?
- How can you prepare for the post-quantum future, today?
- What does post-quantum readiness look like across technology service providers?



Disclaimers

Preparing for a quantum-safe future

- Mathematics free
- Computer science free(ish)
- Quantum mechanics free
- Physics free
- Post-quantum attacks are hypothetical
- Mileage may vary

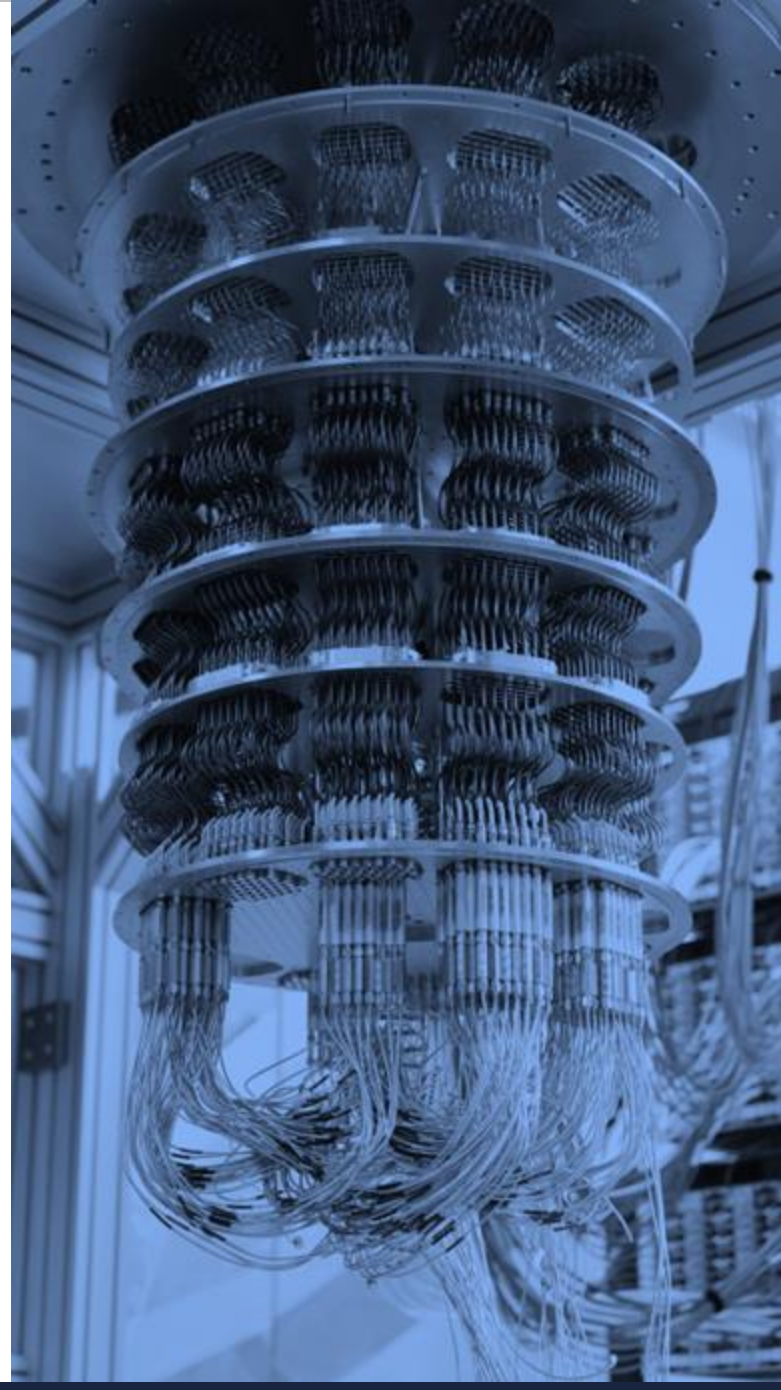


What is Quantum Computing?

What does it mean for the Enterprise?

- Quantum computing is the multidisciplinary field working towards the development and production of quantum computers
- Quantum computers leverage quantum mechanics to solve certain types of problems faster than classical computers
- Quantum computers have the potential to revolutionise and transform certain computational workloads including simulations and data analysis

- Quantum computers also have the potential to solve (break) encryption which relies on problems which cannot be solved in a timely way by classical computers today

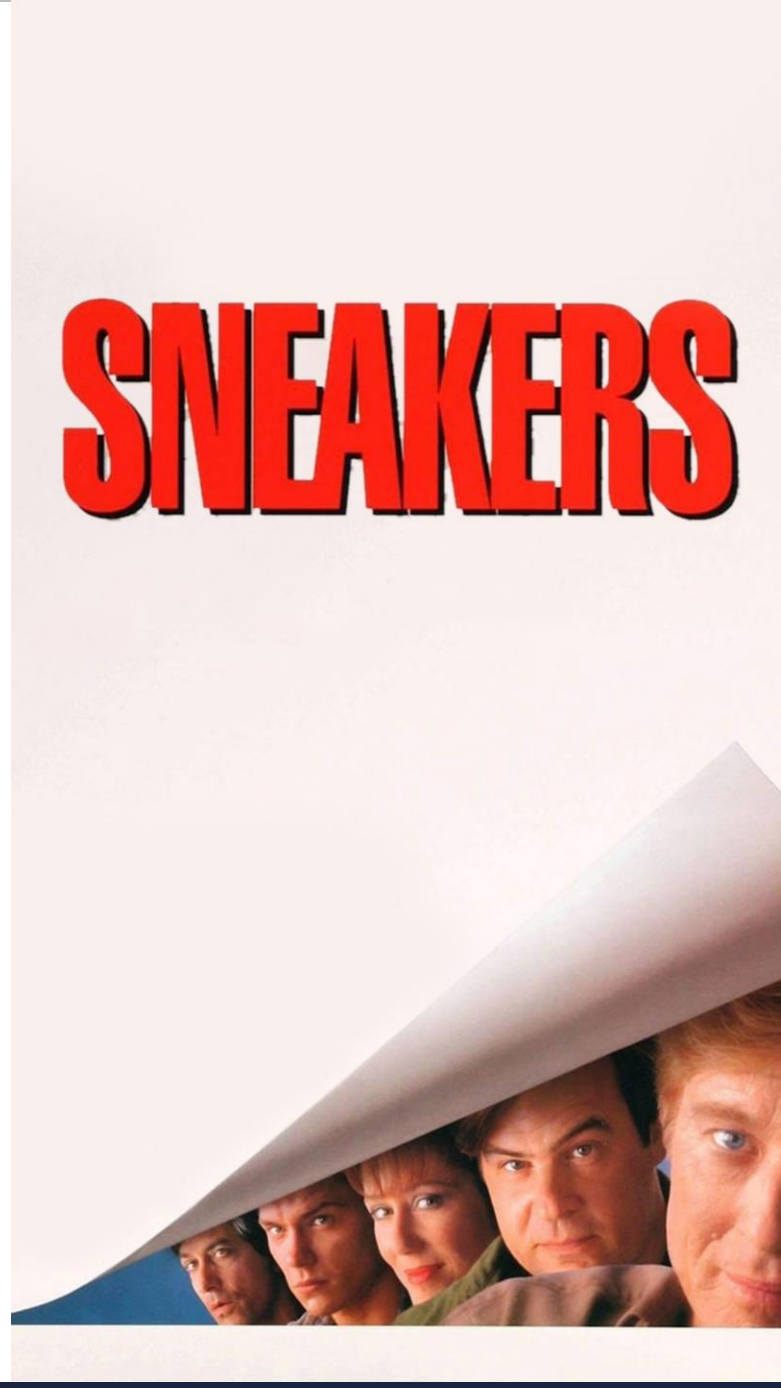


What is Quantum Computing?

What does it mean for the Enterprise?

Sneakers (1992) / Synopsis

"Computer hacker Martin (Robert Redford) heads a group of specialists who test the security of various San Francisco companies. Martin is approached by two National Security Agency officers who ask him to steal a newly invented decoder. Martin and his team discover that ***the black box can crack any encryption code****, posing a huge threat if it lands in the wrong hands."*



What are the risks and how serious is the threat from quantum computing?

- Classical cryptography is used to secure modern cloud, infrastructure, networks, hardware and software

  - **Broken Confidentiality –** encryption prevents unauthorised access to information at rest, in transit and during processing
  - **Broken Integrity –** we use encryption to verify information and identify tampering
  - **Broken Authentication –** encryption underpins all modern authentication
  - **Broken Compliance –** the use of encryption underwrites all security frameworks and standards
  - **Broken Trust –** we rely on encryption to alleviate concerns which would otherwise make the use of public cloud, SaaS and even the public internet impossible

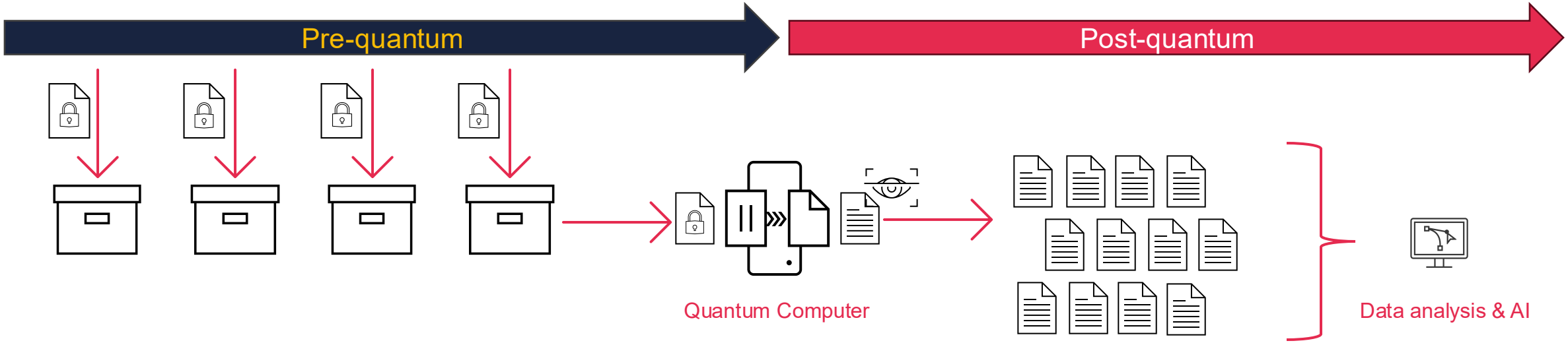


What are the risks and how serious is the threat from quantum computing?

- We can no longer treat a post-quantum future as a purely hypothetical concern
- Preparation will take significant time, planning and effort
- Advances in hardware and software development only shorten our time to prepare for scalable quantum computing
- Grants and funding allocations signal increased concern at a national level
- The post-quantum readiness narrative is consistent across cloud providers
- **Quantum computers are now on a realistic timeline (years/decade)**
  - We don't know exactly when scalable quantum computing will exist (or if it does already)
  - We don't know who (friend or foe) will have access to quantum computing capabilities
  - We don't know to what extent **harvest now, decrypt later** has been a goal/objective

Harvest Now, Decrypt later

Long-term adversarial thinking



2025 Key Trends in Security & Timeline for Postquantum Cryptography

2025 Key Trends in Security

Strategic opportunities and risks from AI and geopolitical change will drive security

- Adapt security to be responsive to current and emerging geopolitical risks
- Define governance, monitor malicious usage and evaluate defensive capabilities of AI
- Prepare for disruptions as the security vendor landscape shifts and consolidates

Architectural patterns will enable security and resilience

- Use "security by design" principles
- Make zero-trust principles' key to security architecture
- Implement CSMA principles starting with data analytics and risk modeling
- Adopt security service edge to replace stand-alone network and cloud access solutions

Software supply chain security will become a critical tenet of application security

- Secure software supply chains
- Perform threat modeling on applications and APIs to support secure-by-design initiatives
- Address application workload and control plane security in the cloud with CNAPP
- Explore emerging penetration testing options to support secure application and API testing

SecOps with AI and au… will enhance ca…

- Evaluat… enh…

Data-centric security will drive modern business initiatives

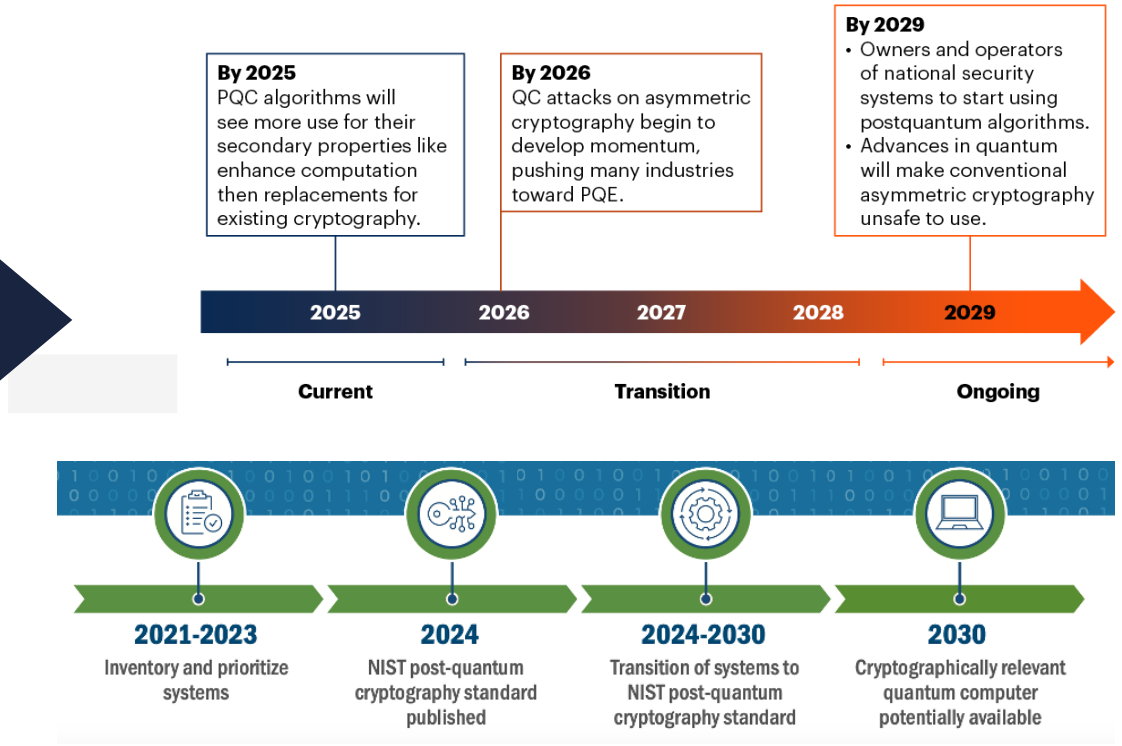
- Gain insights into data with data discovery, classification, tagging and mapping
- Protect sensitive information in data and analytics

…ice will be a focus across …corporate resources

- Focus on resilience to protect against highly disruptive incidents
- Treat mobile security as a critical element of your enterprise attack surface
- Align the goals of the desktop engineering team with those of the security organization

- Prepare for postquantum cryptography disruptions with crypto-agility

Timeline for Postquantum Cryptography



Gartner®

How can you prepare for the post-quantum future?

Post-quantum encryption and Crypto-agility

Start discussing and planning for post-quantum today

- Introduce the topic as a developing concern, establish a post-quantum *working group*
- Start speaking to critical suppliers about their post-quantum readiness plans
  - Public cloud and SaaS
  - Managed service providers
  - SASE providers and ISPs
  - Identity providers (IDPs)
  - Backup and BC/DR providers
- Discuss post-quantum roadmaps with vendors (shared responsibility = shared fate)
- Incorporate post-quantum questions into procurement and renewals
- Begin to prepare for postquantum cryptography disruptions with crypto-agility
- Reflect on client requirements or assurance efforts which have included significant focus on the use of encryption, cloud services or both
- Incorporate quantum vulnerability assessments as a testing objective for all forms testing

Inventory

- Inventory the most sensitive data and information that must be secured
- Inventory all the systems using cryptographic technologies for any function
- Identify where and for what purpose public key cryptography is being used
- Forming a view of real-world **cryptographic agility**
- Leverage existing inventory capabilities including;
  - Vulnerability management and SAST
  - Asset management
  - ID&A and PAM
  - EDR
  - Experience monitoring

What does post-quantum readiness look like across technology service providers?

Provider	Activity	Links	Updated
Microsoft	Updating Symcrypt with PQE Supporting IETF, NIST, OpenQuantum	Starting your journey to become quantum-safe Approach to quantum-resistance	Dec 2024
Google	Early adoption and implementation of emerging standards Supporting IETF, NIST, ISO	PQC at Google PQC Threat model How Google is preparing for a post-quantum world	Aug 2024
AWS	Early adoption of new IETF standards	AWS post-quantum migration plan	Dec 2024
Apple	Protected iMessage with E2EE	iMessage with PQ3	Feb 2024
Zoom	Protected Workplace with E2EE	Zoom Workplace with Kyber 768/FIPS-203	May 2024
Meta	Updating Fizz TLS with PQC	Post-quantum TLS	May 2024
DropBox	Preparing to support PQC	End-to-end encryption for Dropbox	Jun 2024

Useful Links

Preparing for a quantum-safe future, Securing the Law Firm 2025

- Amazon Web Services: What is Quantum Computing?
- Microsoft Azure: What is Quantum Computing?
- Gartner: What Is Quantum Computing? And Why Executives Should Care
- Google Cloud Security Podcast: EP164 Quantum Computing: Understanding the (very serious) Threat and Post-Quantum Cryptography
- Google Security Blog: Post-Quantum Cryptography: Standards and Progress
- Google Bug Hunters Blog: Google's Threat model for Post-Quantum Cryptography
- Google Bug Hunters Blog: Cryptographic Agility and key Rotation
- CISA & NIST: Quantum-readiness: Migration to post-quantum cryptography
- NIST: What is Post-quantum Cryptography?
- NIST: NIST Releases First 3 finalized post-quantum encryption standards
- DHS: Post-quantum Cryptography
- NCSC: Next steps in preparing for post-quantum cryptography

